

AMDOX TECHNOLOGIES

Enterprise AI-Powered Cloud ERP Suite

Next-Generation Intelligent Resource Planning Platform

Industry Project • Amdox Technologies • April 2026

Project Code: AMX-ERP-2026-04 | Version 1.0

Field	Details	Field	Details
Project Title	AI-Powered Cloud ERP Suite	Version	1.0
Organization	Amdox Technologies	Date	April 2026
Domain	Enterprise Software / Cloud Infrastructure	Project Code	AMX-ERP-2026-04
Prepared For	Amdox Technologies – Engineering Division	Classification	Internal

Executive Summary

A scalable, AI-augmented, multi-tenant ERP platform delivering financial management, supply chain automation, HR & payroll, project tracking, and business intelligence — purpose-built for mid-market and enterprise organisations operating across geographies. The platform integrates machine learning models for demand forecasting, anomaly detection, and intelligent approval workflows, while meeting enterprise-grade standards for security, compliance, and reliability.

Target SLA: 99.9% uptime | < 300ms P95 API latency | SOC 2 Type II alignment | GDPR & ISO 27001 compliant architecture

1. Project Overview & Business Value

1.1 Objectives

Deliver a cloud-native ERP suite that consolidates fragmented enterprise workflows into a single intelligent platform, reducing operational overhead, enabling data-driven decisions, and accelerating time-to-insight for business leaders.

1.2 Target Users

User Segment	Primary Use Case
C-Suite / Executives	Real-time dashboards, KPI monitoring, board-level reporting
Finance Teams	GL management, AP/AR automation, multi-currency reconciliation

HR & Payroll Teams	Employee lifecycle, attendance, payroll processing, compliance
Supply Chain Managers	Procurement, inventory, vendor management, demand planning
Project Managers	Resource allocation, milestone tracking, budget management
IT Administrators	Tenant configuration, SSO, audit logs, security policies

1.3 Business Value Delivered

Value Driver	Expected Impact
Unified Data Platform	Eliminate data silos across 6+ legacy systems
AI-Driven Forecasting	Reduce inventory carrying cost by 15–25%
Automated Approvals	Cut approval cycle time from days to minutes
Self-Serve BI	Reduce ad-hoc reporting requests by 60%
Multi-Tenant SaaS Model	Enable Amdox to productize for enterprise clients

1.4 Non-Functional Requirements

NFR Category	Target	Measurement Method
Availability	99.9% monthly uptime (excluding planned maintenance)	UptimeRobot + PagerDuty
API Latency	< 300ms P95 for all REST endpoints	Prometheus + Grafana
Throughput	>= 2,000 concurrent active users per tenant	k6 load test
Data Durability	RPO < 15 min, RTO < 60 min	DR runbook validation
Security	OWASP Top 10 2021 + SOC 2 controls	Quarterly pen-test
Scalability	Horizontal auto-scale on >70% CPU/memory	Kubernetes HPA

2. Detailed Functional Requirements

ID	Module / Feature	Description	Acceptance Criteria
F-01	Multi-Tenant Auth (SSO)	SAML 2.0 / OIDC integration with Azure AD, Google Workspace; MFA enforcement per tenant	Login < 2s; MFA enforced; tenant isolation verified
F-02	Financial Ledger (GL)	Double-entry accounting, multi-currency, period close, intercompany transfers	Zero unbalanced entries; FX rates auto-fetched; period lock enforced
F-03	AP / AR Automation	Invoice OCR, 3-way matching, payment runs, aging reports	OCR accuracy >= 95%; matching auto-approves in < 30s

F-04	HR & Payroll Engine	Employee onboarding, leave management, payroll calculation, statutory compliance	Payroll processed in < 5 min for 10k employees; audit trail complete
F-05	Supply Chain & Inventory	PO lifecycle, vendor portal, real-time stock levels, reorder automation	Reorder triggered at configurable threshold; vendor notified via email/webhook
F-06	AI Demand Forecasting	ML model for SKU-level demand prediction (LSTM / Prophet)	MAPE < 12% on 90-day horizon; model retrain weekly
F-07	Project Management	Gantt, resource allocation, budget tracking, milestone alerts	Overrun alert when actual > budget by 10%; Gantt renders < 1s
F-08	Business Intelligence	Drag-and-drop dashboard builder, scheduled reports, drill-down analytics	Dashboard saved in < 500ms; exports to PDF/Excel
F-09	Audit & Compliance Log	Immutable audit trail for all mutations; GDPR data subject requests	Tamper-evident logs; DSR fulfilled in < 72h
F-10	Notification Engine	In-app, email, SMS, webhook for configurable business events	Delivery confirmed; retry up to 3x on failure; channel preference per user
F-11	API Gateway & Webhooks	REST + GraphQL gateway; outbound webhook subscriptions	OpenAPI 3.1 spec published; all endpoints versioned
F-12	Offline / PWA Support	Service worker cache for critical read views; sync on reconnect	Core views functional offline; sync completes on reconnect without data loss

3. Technology Stack – Production Grade 2026

Category	Selected Technology	Rationale / Alternatives Considered
Frontend Framework	Next.js 15 + React 19 + TypeScript 5.5	SSR/SSG for fast initial load; App Router; Remix & Vite evaluated
UI Component Library	shadcn/ui + Radix + Tailwind CSS 4	Accessible, unstyled primitives; MUI too opinionated for custom brand
State Management	Zustand + React Query (TanStack v5)	Lightweight; React Query handles server state, Zustand for client state
Data Visualisation	Recharts + ECharts (Apache) + D3.js	Recharts for dashboards; ECharts for heavy analytics; D3 for custom viz
Backend Runtime	Node.js 22 LTS + TypeScript 5.5	Bun evaluated; Node chosen for ecosystem maturity
API Framework	NestJS 11 (modular monolith)	Fastify evaluated; NestJS DI + decorators ideal for large domain

API Protocols	REST (OpenAPI 3.1) + GraphQL (Apollo v4)	GraphQL for flexible BI queries; REST for CRUD and integrations
Primary Database	PostgreSQL 17 + Prisma ORM	ACID compliance critical for ERP; MongoDB considered then ruled out
Time-Series DB	TimescaleDB (extension)	Audit logs and telemetry; InfluxDB evaluated
Cache & Session	Redis 8 (Dragonfly-compatible) + ioredis	Session store, job queues, real-time pub/sub
Message Queue	BullMQ (Redis-backed)	Payroll jobs, email, webhooks; Kafka for future high-throughput scale
AI / ML Services	Python 3.13 + FastAPI + scikit-learn + Prophet	Dedicated ML microservice; TensorFlow/PyTorch for LSTM model
Search	Elasticsearch 8.15 / OpenSearch	Full-text vendor/product/document search; Typesense evaluated
File Storage	AWS S3 + CloudFront (or MinIO self-hosted)	Invoice attachments, exports, backups
Authentication	Keycloak 25 (OIDC/SAML) + JWT (RS256)	Enterprise SSO; Auth0 evaluated; Keycloak for on-prem flexibility
Email Delivery	AWS SES + Resend fallback	High deliverability; DKIM/SPF configured
Containerisation	Docker 27 multi-stage + Distroless base	Minimal attack surface; Podman compatible
Orchestration	Kubernetes 1.31 + Helm 3 charts	EKS / GKE deployment; ArgoCD GitOps
Service Mesh	Istio 1.22 (mTLS + traffic policies)	Zero-trust networking; Linkerd evaluated
CI/CD	GitHub Actions + ArgoCD	Matrix builds; progressive delivery; GitLab CI alternative
Observability	OpenTelemetry + Prometheus + Grafana + Loki	Distributed tracing + metrics + logs; Datadog evaluated
Security Scanning	Trivy + Snyk + OWASP ZAP	SAST/DAST/container scanning in pipeline
Testing	Vitest + Playwright + k6 + Jest	Unit/integration/E2E/load coverage
IaC	Terraform 1.9 + Terragrunt	AWS + GCP multi-cloud; Pulumi evaluated

4. Granular 28-Day Execution Plan

Week 1 – Foundation, Architecture & Authentication

Day 1 – Discovery & Planning

- Stakeholder requirements workshop (finance, HR, supply chain, IT)
- Competitor analysis matrix: SAP S/4HANA, Oracle NetSuite, Microsoft Dynamics 365, Odoo 17
- User story mapping + acceptance criteria in Notion / Jira
- Define MVP feature scope and de-scope list

Day 2 – Architecture Design

- C4 diagrams: Context, Container, Component (Excalidraw / Structurizr)
- Database ERD: Tenant, User, Role, Account, Transaction, JournalEntry, Employee, PurchaseOrder, InventoryItem, Notification

- API contract design (OpenAPI 3.1 spec skeleton)
- Decide on monorepo structure (Turborepo + pnpm workspaces)

Day 3 – Repository & DevOps Skeleton

- Turborepo monorepo: apps/web, apps/api, apps/ml-service, packages/ui, packages/db
- ESLint + Prettier + Husky + lint-staged + commitlint (conventional commits)
- Docker Compose local dev stack: Postgres, Redis, Keycloak, Elasticsearch
- .env.example + secret management strategy (Vault / AWS Secrets Manager)

Day 4 – Authentication & Multi-Tenancy

- Keycloak realm setup: OIDC configuration, SAML adapter, realm-per-tenant strategy
- NestJS auth module: passport-jwt, RBAC guard, tenant context middleware
- User table + TenantUser join with roles (SuperAdmin, TenantAdmin, Manager, Viewer)
- Refresh token rotation + blacklist (Redis SET)

Day 5 – Core Domain Models & Database

- Prisma schema: all core entities with indexes, soft-delete, created_at/updated_at
- Row-level security strategy: tenantId filter injected at query layer
- Database migrations pipeline + seeder scripts
- Prisma Studio setup for internal dev inspection

Day 6 – API Gateway & Base Endpoints

- NestJS app bootstrap: global exception filters, validation pipe (class-validator), interceptors
- Health check endpoints: /health/live, /health/ready, /health/db
- OpenAPI swagger UI served at /api-docs
- Postman collection auto-generated from OpenAPI spec

Day 7 – Week 1 Review & Demo

- Internal demo: auth flow, multi-tenant isolation, API docs
- Fix any schema or auth edge cases
- Write ADRs (Architecture Decision Records) for key choices

Week 2 – Finance, HR & Supply Chain Modules

Day 8–9 – Financial Ledger & AP/AR

- General Ledger: chart of accounts, journal entries, double-entry validation
- Multi-currency support: ECB/OpenExchangeRates daily FX fetch + historical rates
- AP module: invoice creation, 3-way PO/GR/Invoice matching, payment run logic
- AR module: customer invoicing, payment recording, aging report query
- Period close locking mechanism with role-based override

Day 10 – HR Core & Employee Lifecycle

- Employee CRUD: personal info, employment contract, department, reporting hierarchy
- Organisational chart (recursive CTE query in Postgres)
- Leave management: leave types, accrual rules, approval workflow (state machine)
- Attendance module: clock-in/out API, overtime calculation

Day 11 – Payroll Engine

- Gross-to-net payroll calculation engine (configurable tax slabs, statutory deductions)
- Payroll run: batch processing via BullMQ (async, retry-safe)
- Payslip PDF generation (Puppeteer / pdfkit)
- Audit trail for every payroll mutation

Day 12–13 – Supply Chain & Inventory

- Purchase requisition → PO → goods receipt workflow
- Vendor master data management + vendor portal API
- Inventory: real-time stock levels, FIFO/FIFO costing, warehouse locations
- Reorder point automation: trigger PO draft when stock < threshold
- Supplier email notification via BullMQ + AWS SES

Day 14 – Integration Testing & Bug Fixes

- Integration tests for all Week 2 modules (Vitest + Supertest)
- Database query optimisation (EXPLAIN ANALYZE on critical queries)
- Cross-module smoke test: PO → inventory → AP invoice → GL journal entry

Week 3 – AI/ML, BI Dashboard, Security & Notifications

Day 15–16 – AI Demand Forecasting Microservice

- Python FastAPI ML service: /train, /predict, /health endpoints
- Prophet model for time-series SKU demand forecasting
- LSTM model (PyTorch) as secondary model for high-volume SKUs
- Model versioning with MLflow (or simple file-based versioning)
- Weekly retraining job scheduled via BullMQ + Redis cron
- NestJS forecasting module: consume ML service, cache predictions in Redis

Day 17 – Business Intelligence Dashboard

- Dashboard builder backend: widget configuration stored as JSON in Postgres
- Recharts + ECharts React components: bar, line, pie, heatmap, funnel
- Drill-down: click chart segment → filtered data table
- Scheduled report job: generate PDF/Excel + email delivery
- Real-time metric refresh via Server-Sent Events (SSE)

Day 18 – Project Management Module

- Project CRUD: milestones, tasks, dependencies (DAG validation)
- Gantt chart rendering (react-gantt or custom D3 implementation)
- Resource allocation: assign employees to tasks, utilisation heatmap
- Budget tracking: planned vs actual with variance alerts

Day 19 – Notification & Event Engine

- Event bus: domain events emitted via NestJS EventEmitter2
- Notification service: in-app (SSE / Socket.io), email (SES), webhook (signed HMAC)
- User notification preferences per channel and event type
- BullMQ dead-letter queue + retry dashboard (Bull Board UI)

Day 20 – Security Hardening

- OWASP Top 10 checklist: CSRF, XSS (DOMPurify + CSP headers), IDOR checks
- Helmet.js: HSTS, X-Frame-Options, Referrer-Policy, Permissions-Policy
- Rate limiting: redis-based sliding window (nestjs-throttler)
- Input validation: class-validator on all DTOs; Zod on frontend forms
- Secrets rotation plan + no committed secrets audit (trufflehog scan)
- Trivy container scan + Snyk dependency audit integrated in CI

Day 21 – Performance & Load Testing

- k6 load test: 2,000 concurrent virtual users, 10-minute steady state
- Identify bottlenecks: N+1 queries, missing indexes, Redis cache gaps
- Postgres read replicas strategy for BI/reporting queries
- React frontend: Lighthouse score >= 90 (performance, accessibility)

- Bundle analysis: webpack-bundle-analyzer, code splitting strategy

Week 4 – Deployment, Observability & Final Polish

Day 22 – Containerisation

- Multi-stage Dockerfiles for all services (Distroless final stage)
- docker-compose.prod.yml with health checks and resource limits
- Container image scanning with Trivy in CI pipeline
- .dockerignore optimisation for minimal build context

Day 23 – Kubernetes Manifests & Helm Charts

- Helm chart: Deployment, Service, ConfigMap, Secret (Sealed Secrets), Ingress, HPA, PDB
- Istio virtual services + destination rules for canary deployment
- Namespace isolation per environment (dev / staging / prod)
- minikube / kind local cluster validation

Day 24 – CI/CD Pipeline

- GitHub Actions matrix: lint → unit test → integration test → build → docker push → deploy
- ArgoCD application manifest for GitOps continuous delivery
- Smoke test suite triggered post-deployment
- Slack notifications for pipeline success/failure

Day 25 – Cloud Deployment

- Frontend: Vercel (Next.js) or AWS CloudFront + S3
- Backend: AWS EKS / Railway / Fly.io (staging first, then prod)
- Postgres: RDS Aurora Serverless v2 / Supabase
- Redis: ElastiCache / Upstash
- Custom domain + Let's Encrypt TLS (cert-manager)
- DNS / CDN configuration + WAF rules

Day 26 – Observability Stack

- OpenTelemetry SDK instrumentation (traces, metrics, logs) across all services
- Prometheus scrape configs + Grafana dashboards (latency, error rate, saturation)
- Loki log aggregation + Grafana log explorer
- PagerDuty / OpsGenie alerting for SLA breaches
- Distributed trace sampling: 100% errors, 10% success

Day 27 – Demo Video & Documentation

- Record 5–7 min scenario-based demo (multi-user, multi-module walkthrough)
- Architecture diagram screenshot + deployment topology diagram
- README.md: local setup, env vars, architecture, screenshots, video link
- Export project report to PDF (this document)

Day 28 – Final QA & Submission

- Cross-browser QA: Chrome, Firefox, Safari, Edge
- Mobile responsive validation (375px, 768px, 1440px breakpoints)
- Accessibility audit: ARIA labels, keyboard nav, colour contrast (WCAG 2.1 AA)
- Final Lighthouse + k6 report screenshots
- Submission package: PDF report + GitHub repo + live demo URL + demo video

5. Architecture Overview

5.1 System Layers

Layer	Components	Communication Pattern
Client Layer	Next.js 15 SPA/SSR, React Native (mobile roadmap)	HTTPS REST + GraphQL + SSE
API Gateway	NestJS Gateway, rate limiter, auth middleware, request logging	JWT validation + tenant context injection
Core Services	Finance, HR, Supply Chain, Project, Notification services	In-process modules (monolith) → future microservices
ML Service	Python FastAPI, Prophet + LSTM, MLflow model registry	Internal REST over Istio mTLS
Data Layer	PostgreSQL 17 + TimescaleDB, Redis 8, Elasticsearch 8	Prisma ORM, ioredis, Elasticsearch client
Infrastructure	Kubernetes 1.31, Istio, ArgoCD, Terraform	GitOps + IaC
Observability	OpenTelemetry, Prometheus, Grafana, Loki, PagerDuty	Pull-based metrics + push-based traces

5.2 Key Design Patterns

Pattern	Application in This System
Domain-Driven Design (DDD)	Bounded contexts per module (Finance, HR, SCM); Aggregate roots enforce invariants
CQRS (lightweight)	Read models (optimised SQL views) separate from write models for BI queries
Outbox Pattern	Guaranteed event delivery: DB transaction + outbox table consumed by BullMQ worker
Saga (Orchestration)	Payroll run: multi-step saga with compensating transactions on failure
Repository Pattern	All database access via Prisma repositories; no raw SQL outside reporting layer
Decorator / Guard (NestJS)	RBAC, tenant isolation, audit logging via declarative decorators

6. Security & Compliance Framework

Domain	Controls Implemented	Standard / Reference
Identity & Access	MFA, RBAC, ABAC, Just-in-time access, session timeout	NIST 800-63 / Zero Trust
Data Encryption	TLS 1.3 in transit, AES-256 at rest (RDS encryption, S3 SSE)	FIPS 140-2 aligned
Input Validation	Server-side DTO validation + frontend Zod schemas + SQL injection prevention via Prisma parameterised queries	OWASP A03:2021

API Security	CORS allowlist, rate limiting (sliding window), HMAC webhook signatures, API versioning	OWASP API Security Top 10
Dependency Security	Snyk SCA scans in CI, Trivy container scanning, Renovate bot for auto-updates	CIS Benchmarks
Audit Logging	Immutable audit log (TimescaleDB append-only), tamper detection via hash chaining	SOC 2 CC7.2
GDPR Compliance	Data subject request API, consent management, right to erasure soft-delete pipeline	GDPR Art. 17, 20
Secrets Management	HashiCorp Vault / AWS Secrets Manager; Sealed Secrets in K8s; zero hardcoded secrets	CIS K8s Benchmark

7. Deployment & Operations

7.1 Environment Strategy

Environment	Purpose	Deployment Trigger	Data Policy
Development	Local feature development; Compose stack	Manual / on-save hot reload	Synthetic seed data
Staging	Integration QA, UAT, load testing	Merge to main branch (ArgoCD)	Anonymised prod snapshot
Production	Live customer traffic	Tagged release + manual approval gate	Real data; GDPR controls active

7.2 Runbook Highlights

Scenario	Procedure
DB Failover	Aurora promotes read replica automatically; health check flips DNS; RTO < 5 min
Pod OOM / CrashLoop	Liveness probe triggers restart; alert fires after 3 restarts; HPA scales out
Cache Miss Storm	Circuit breaker in Redis client; fallback to DB with adaptive TTL warmup
Payroll Job Failure	Saga compensates: revert partial calculations; send alert; manual retry endpoint
Security Incident	WAF blocks IP; alert to PagerDuty; Vault rotates affected secrets within 15 min

8. Evaluation Criteria

Category	Points	Focus Areas
Innovation & Problem Solving	15	AI forecasting originality, novel UX patterns, thoughtful architecture decisions
Technical Depth & Best Practices	25	Clean code, DDD, security hardening, OWASP compliance, design patterns
Functionality & User Experience	20	All F-01–F-12 met, responsive design, accessibility (WCAG 2.1 AA)
Documentation Quality	20	Well-structured report, ADRs, API docs (OpenAPI), README
Deployment & Reliability	10	Stable live demo, CI/CD pipeline, monitoring in place
Presentation & Polish	10	Demo video quality, visual appeal, scenario depth
TOTAL	100	

9. Submission Guidelines

#	Deliverable	Format / Location	Required	Weight
1	Project Report (this document)	1 PDF, A4, 8–15 pages	Yes	25%
2	Live Public Demo URL	HTTPS, no VPN required	Yes	30%
3	GitHub Repository	Public repo, clear folder structure	Yes	20%
4	README.md	Setup, architecture, screenshots, video link	Yes	15%
5	Demo Video	5–7 min, YouTube Unlisted / Loom / Drive	Yes	10%

Naming convention: YourName_AMX_ERP_AmdoxTechnologies_April2026.pdf / .zip

Submit via the Amdox Technologies project dashboard — no email submissions accepted.

Strict deadline enforced. Late submissions are automatically disqualified from stipend consideration.

Crafted with precision and modern engineering principles
Amdox Technologies • Engineering Division • April 2026

